Autonomous Agent Tooling: An Empirical Review

Abstract: This paper presents an investigation into low-overhead document processors for multi-agent daemon systems.

Modern autonomous agent architectures face strict resource bounds. When agents execute in multi-tenant background daemons, launching full browser runtimes for routine document transformations imposes high latency penalties and memory overhead.

By contrasting headless browser architectures with lightweight native implementations, we identify optimal operating regimes. Specialized layout algorithms achieve sub-millisecond execution times while bounding resident set sizes.

Furthermore, reading pipelines require careful handling of text stream extraction. Naive tokenizers frequently fail on CIDFont subsets, whereas robust AST interpreters reliably recover structural semantics.

Experimental evaluations confirm that browserless rendering satisfies print media fidelity while conserving up to 90% of runtime host memory.